

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1000

Glenbrook 138 kV
March 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1000. The Interconnection Customer proposes a 446.1 MW natural gas combined cycle facility interconnecting at the Glenbrook 138 kV substation in the West Virginia area. The request was received on May 23, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 446 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$109,211,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Arvin / Shuffleton Substation 138 kV line	Clark County / Kearneyville 345 kV	188.9	223	22.6
Palmyra / Grimmway-Malaga Substation 138 kV	Clark County / Kearneyville 345 kV Line	188.2	313	53.7
Glenbrook / Palyra 138 kV Line	Clark County / Kearneyville 345 kV	185.1	325	53.7
Arvin / Garden Corners 138 kV	Douglas City / Southwest Research Institute 138 kV	173.4	199	20.6
Southwest Research Institute / Arvin 138 kV L	Garden Crners / Douglas City 138 kV	172.4	226	22.9
Claytonia Substation / Shuffleton 138 kV	Clark County / Kearneyville 345 kV	163.4	241	22.6
Douglas City / Southwest Research Institute 138 kV	Garden Corners / Douglas City 138 kV	161.2	261	22.9
Garden Corners / Douglas City 138 kV	Southwest Research Institute / Douglas City 138 kV	158.2	241	20.6
Smallwood / Killingsworth 138 kV	Palmra / Glenbrook 138 kV Line	152.8	221	58.1
Millville Substation / Douglas City 138 kV Line	Clark County / Kearneyville 345 kV	147.1	461	27.0
Millville / Sleepy Hollow 138 kV	Clark County / Kearneyville 345 kV	146.0	474	27.0
Sun City Substation / Peachland 138 kV	Kearneyville / Clark County 345 kV	145.2	355	10.8
Grimmway-Malaga / Sleepy Hollow 138 kV	Kearneyville / Clark County 345 kV L	144.8	487	27.0
Columbus City / Smallwood 138 kV Line	Glenbrook / Palmyra 138 kV	140.0	231	58.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Arvin / Carlota 138 kV	Arvin Substation / Shuffleton 138 kV	136.8	198	18.0
Lamont / University City 138 kV Line	Clark County / Kearneyville 345 kV Line	134.0	342	10.8
Air Force Base 345/138 kV Transformer	Tiger Bay / Air Force Base 345 kV	134.0	541	6.4
University City / Richford 138 kV	Kearneyville / Clark County 345 kV	129.3	381	10.8
Redman / Glenbrook 138 kV	Glenbrook Substation / Palmyra 138 kV	129.2	336	100.0
Richford / Sun City 138 kV	Kearneyville / Clark County 345 kV line	127.6	397	10.8
Clark County / Kearneyville 345 kV	Grimmway-Malaga Substation / Sleepy Hollow 138 kV	127.4	1,383	25.4
Gene Substation / Grimmway-Malaga 138 kV	Glenbrook / Palmyra 138 kV	124.1	314	7.4
Gallegos / Redman 138 kV	Palmyra / Glenbrook 138 kV	124.1	344	100.0
Tanner / Gene Substation 138 kV Line	Glenbrook / Palmyra 138 kV Line	123.3	327	7.4
Tanner / Lcec No. 5 138 kV Line	Glenbrook / Palmyra 138 kV	122.4	336	7.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Arvin Substation / Shuffleton 138 kV	Rebuild line with higher-capacity conductor	\$2,114,000	82% (\$1,726,000)
Palmyra / Grimmway-Malaga 138 kV	Rebuild line with higher-capacity conductor	\$1,763,000	100% (\$1,763,000)
Palmyra / Glenbrook 138 kV Line	Rebuild line with higher-capacity conductor	\$1,345,000	100% (\$1,345,000)
Arvin / Garden Corners 138 kV line	Rebuild line with higher-capacity conductor	\$11,943,000	82% (\$9,755,000)
Arvin / Southwest Research Institute 138 kV L	Rebuild line with higher-capacity conductor	\$7,408,000	82% (\$6,051,000)
Claytonia Substation / Shuffleton 138 kV	Rebuild line with higher-capacity conductor	\$2,547,000	82% (\$2,079,000)
Douglas City / Southwest Research Institute 138 kV	Rebuild line with higher-capacity conductor	\$2,204,000	82% (\$1,800,000)
Douglas City Substation / Garden Corners 138 kV	Rebuild line with higher-capacity conductor	\$2,245,000	82% (\$1,833,000)
Smallwood / Killingsworth Substation 138 kV	Rebuild line with higher-capacity conductor	\$5,006,000	100% (\$5,006,000)
Millville Substation / Douglas City 138 kV Line	Rebuild line with higher-capacity conductor	\$20,166,000	82% (\$16,462,000)
Millville / Sleepy Hollow 138 kV	Rebuild line with higher-capacity conductor	\$5,111,000	82% (\$4,172,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sun City Substation / Peachland 138 kV	Rebuild line with higher-capacity conductor	\$3,029,000	51% (\$1,541,000)
Grimmway-Malaga / Sleepy Hollow 138 kV	Rebuild line with higher-capacity conductor	\$1,701,000	82% (\$1,388,000)
Columbus City / Smallwood 138 kV Line	Rebuild line with higher-capacity conductor	\$2,480,000	100% (\$2,480,000)
Arvin / Carlota 138 kV	Rebuild line with higher-capacity conductor	\$4,543,000	84% (\$3,811,000)
Lamont / University City 138 kV	Rebuild line with higher-capacity conductor	\$11,827,000	51% (\$6,017,000)
Air Force Base 345/138 kV Transformer	Replace transformer with higher-rated unit	\$15,942,000	52% (\$8,222,000)
University City / Richford 138 kV	Rebuild line with higher-capacity conductor	\$5,826,000	51% (\$2,964,000)
Glenbrook / Redman 138 kV Line	Rebuild line with higher-capacity conductor	\$3,858,000	100% (\$3,858,000)
Richford / Sun City 138 kV	Rebuild line with higher-capacity conductor	\$7,835,000	51% (\$3,986,000)
Clark County / Kearneyville 345 kV	Rebuild line with higher-capacity conductor	\$78,633,000	14% (\$11,040,000)
Gene Substation / Grimmway-Malaga 138 kV	Rebuild line with higher-capacity conductor	\$6,888,000	55% (\$3,811,000)
Gallegos / Redman 138 kV	Rebuild line with higher-capacity conductor	\$3,044,000	100% (\$3,044,000)
Tanner / Gene Substation 138 kV Line	Rebuild line with higher-capacity conductor	\$508,000	55% (\$281,000)
Tanner / Lcec No. 5 138 kV Line	Rebuild line with higher-capacity conductor	\$8,633,000	55% (\$4,776,000)
	Total allocated to project		\$109,211,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.